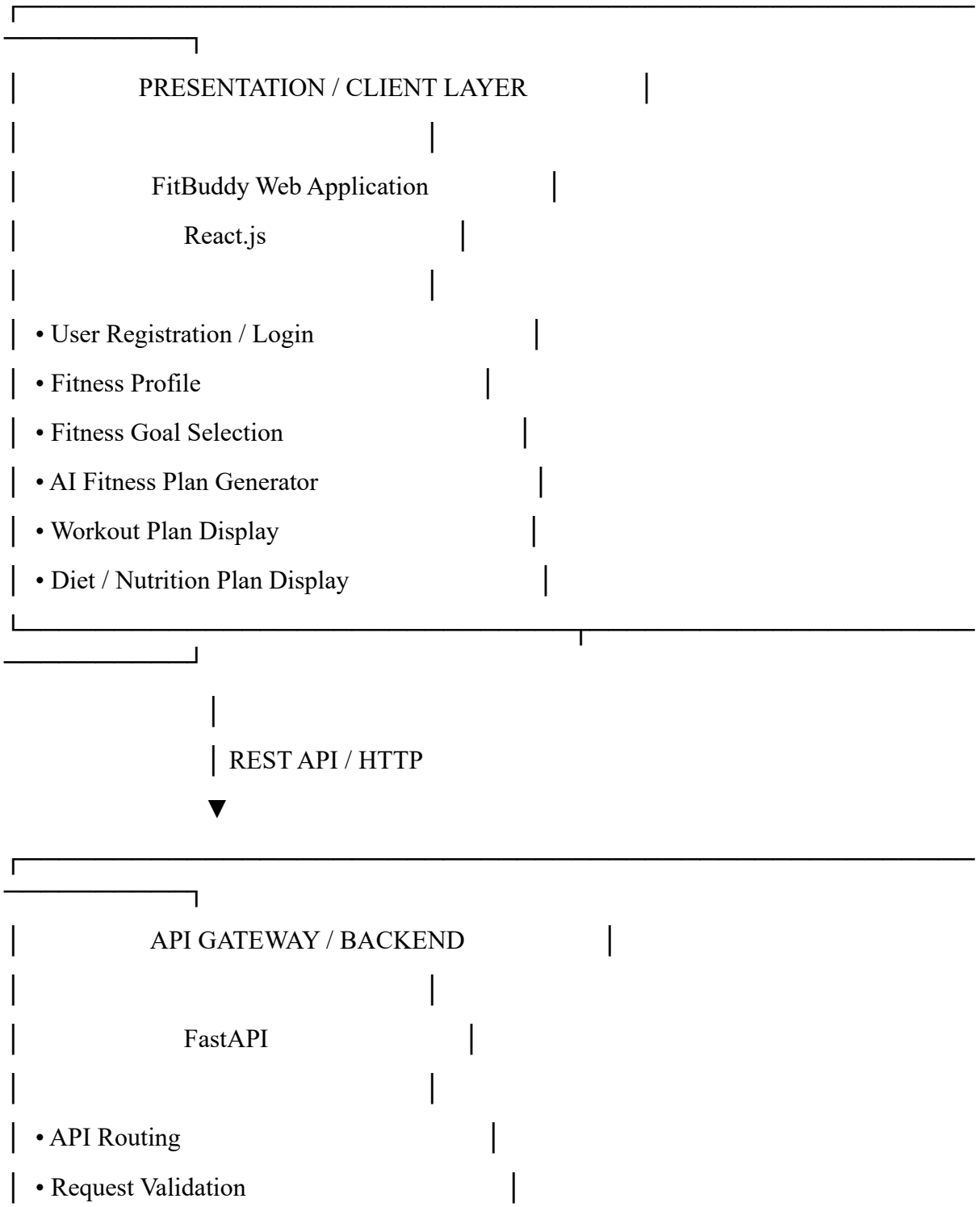
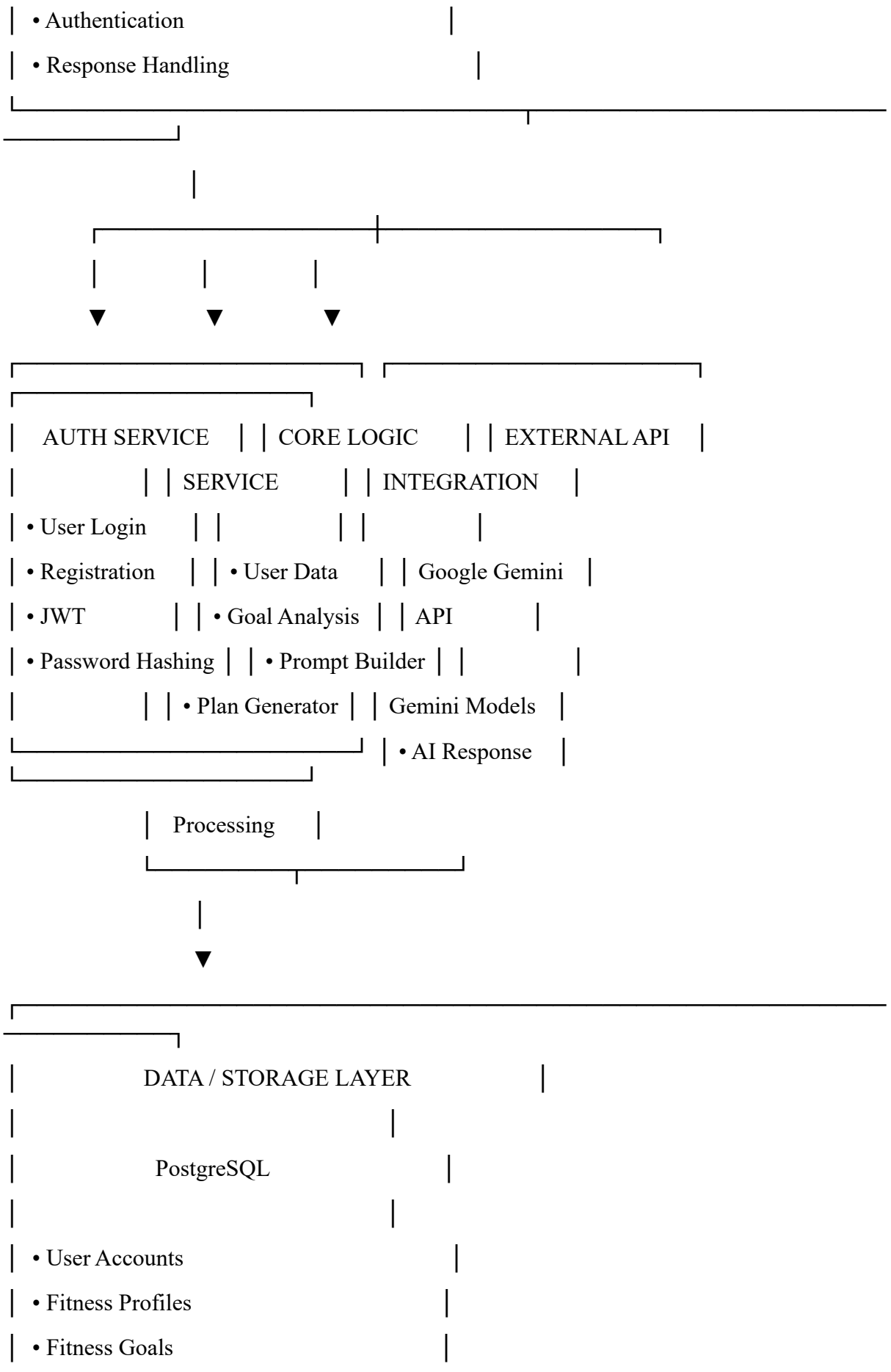
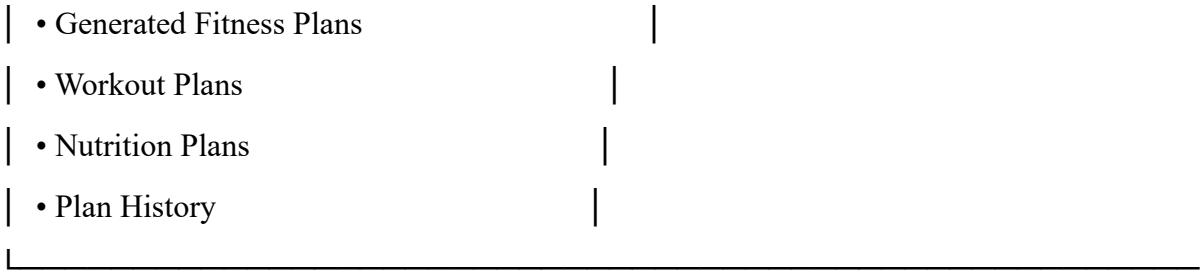
Solution architecture

Date	30 September 2026
Team id	SWTID-2026-2551
Project name	Fit buddy -AI fitness plan generator using gemini models
Maximum marks	5 Marks

Solution Architecture diagram:




• Authentication
• Response Handling
AUTH SERVICE
• User Login
• Registration
• JWT
• Password Hashing
CORE LOGIC SERVICE
• User Data
• Goal Analysis
• Prompt Builder
• Plan Generator
EXTERNAL API INTEGRATION
Google Gemini API
Gemini Models
• AI Response
Processing
DATA / STORAGE LAYER
PostgreSQL
• User Accounts
• Fitness Profiles
• Fitness Goals



Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for registration, login, fitness profile input, goal selection and displaying AI-generated workout and nutrition plans.	React.js, HTML, CSS, JavaScript
API Gateway	Handles communication between the frontend and backend, routes API requests, validates input and manages responses.	Python, FastAPI
Auth Service	Manages user registration, login, password protection and authentication using JWT.	JWT, bcrypt
Core Logic Service	Processes user fitness information, prepares prompts and manages the generation and processing of personalized fitness plans.	Python, FastAPI
Gemini AI Service	Uses Gemini models to generate personalized workout, nutrition and fitness recommendations based on user-provided information.	Google Gemini API, Gemini Models
Database	Stores user information, fitness profiles, goals, generated plans and plan history.	PostgreSQL, SQL Alchemy
External API Integration	Connects the backend application with Google's Gemini API for AI-based fitness plan generation.	Google Gemini API

Fitness Plan Generator	Converts user fitness requirements into structured AI-generated workout and nutrition plans.	Python, Gemini
Plan History	Stores previously generated fitness plans so they can be accessed later.	PostgreSQL

Frontend → React.js

Backend → Python Fast API

Authentication → JWT + b crypt

AI → Google Gemini Models

Database → PostgreSQL

ORM → SQL Alchemy

API → REST API